Customer Retention & Churn Risk Analytics Engine

End-to-End Project Defense Guide | GitHub: github.com/srujanavarsha/customer-retention-analytics

1. The 2-Minute Perfect Project Pitch

'In this project, I addressed a critical e-commerce retention problem: top-of-funnel customer acquisition was growing 18% YoY, but net revenue remained flat due to hidden churn.

To solve this, I designed an end-to-end analytics engine across 1,200 accounts and 5,100+ orders:
*1. **Data Modeling:** Implemented a clean Star Schema in SQLite with a central Fact Orders table and dimension tables.*
*2. **Advanced SQL:** Wrote modular CTEs for monthly cohort retention curves, NTILE-based RFM segmentation, and LAG window functions to detect repurchase cadence anomalies.*
*3. **AI Sentiment Layer:** Used a Python NLP script to classify 2,200 unstructured customer reviews into sentiment scores and operational churn categories.*
*4. **Power BI Cockpit:** Built an executive dashboard with time-intelligence DAX measures and semantic color alerts.*

***The Key Business Result:** We discovered an 81.5% drop-off in Month 1, flagged 276 high-value accounts at risk ($180K revenue exposure), and proved that hardware defects in Electronics drove 52% of negative reviews.'*

2. Data Architecture & Schema Breakdown

You modeled the data using a strict **Star Schema** to avoid bidirectional joins and optimize query speed:

Table	Type	Key Columns & Purpose
`fact_orders`	Central Fact	5,147 rows. Grain: 1 row per order line. Contains `total_amount`, `order_date`, `status`.
`dim_customers`	Dimension (1:*)	1,200 profiles. `customer_id` (PK), `acquisition_channel`, `country`, `customer_type`.
`dim_products`	Dimension (1:*)	22 product items. `product_id` (PK), `category`, `unit_price`, `unit_cost`.
`dim_calendar`	Dimension (1:*)	Created via DAX (`ADDCOLUMNS(CALENDAR(...))`) for time-intelligence filtering.
`customer_feedback_enriched`	Secondary Fact	2,200 reviews enriched with `sentiment` and `churn_driver_category`.

3. The 3 Technical SQL Queries in Your Portfolio

1. Monthly Cohort Decay Matrix (`01_cohort_retention_matrix.sql`):
• Uses CTEs to compute customer's first purchase month (`MIN(order_date)`).
• Calculates `cohort_index` = elapsed calendar months since Month 0.
• Divides returning active buyers by Month 0 cohort size to derive retention %.

2. RFM Customer Segmentation (`02_rfm_customer_segmentation.sql`):
• Computes Recency (days since last order), Frequency (order count), and Monetary (spend).
• Applies `NTILE(4) OVER (...)` across all 3 dimensions to classify users into: *Champions, Loyal, Needs Nurturing, At Risk,*

and Lost.

3. Repurchase Cadence Churn Detection (`03_churn_risk_detection.sql`):
• Uses `LAG(order_date, 1) OVER (PARTITION BY customer_id ORDER BY order_date)` to calculate each customer's personal average order cycle.
• If days since last order exceeds **2.5x their historical average**, SQL flags a *Critical Churn Alert*.

4. The Power BI Executive Dashboard & DAX Measures

• **[Total Net Revenue]:** `CALCULATE(SUM(fact_orders[total_amount]), fact_orders[order_status] = 'Completed')`
• **[Completed Orders]:** `CALCULATE(DISTINCTCOUNT(fact_orders[order_id]), fact_orders[order_status] = 'Completed')`
• **[Active Customers (Last 90 Days)]:** Uses `DATESBETWEEN` on the date dimension to track active accounts.
• **Visual Design Philosophy:** Used card-on-canvas containers, rounded corners (8px), data labels on horizontal bars, and semantic color alerting (Red for hardware defects, Green for positive praise).

5. Top 5 Tough Follow-Up Questions (And How to Answer Them)

Q1: 'Why did you use a Star Schema instead of one big flat CSV table?'

Answer: *'A flat denormalized table creates data redundancy, increases file size, and slows down DAX measures. A Star Schema ensures a clean 1-to-many relationship, optimizes Power BI's columnar engine, and makes time intelligence seamless.'*

Q2: 'Why did you evaluate Churn using personal repurchase cadence instead of a fixed 90-day rule?'

Answer: *'Static thresholds fail because buying behaviors differ. A loyal customer who usually orders every 10 days has likely churned if 40 days pass, while a seasonal buyer ordering every 180 days is completely normal. Personalized cadence anomaly detection with LAG() is industry best practice.'*

Q3: 'How did you classify unstructured reviews into churn categories?'

Answer: *'I built a Python script that processed 2,200 qualitative comments against an operational taxonomy covering Shipping, Defects, Support, and Billing, generating sentiment scores and category tags which were fed directly into Power BI.'*

Q4: 'What was the single most actionable business recommendation you made?'

Answer: *'Because Month 1 drop-off was 81.5%, we recommended an automated 30-day second-purchase incentive for new buyers, as our data proved that customers who place a second order within 45 days generate 4.2x higher 12-month LTV ($820 vs $195).'*

Q5: 'What would you do differently if the dataset was 100 million rows?'

Answer: *'I would migrate from SQLite to an enterprise data warehouse like Snowflake or BigQuery, use dbt for data transformation models, and configure Power BI in DirectQuery or Composite Mode with aggregations.'*